

Introduction to Mathematics and Optimisation

Exercises Week 7

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1 Exercise 5.62

Show from scratch (without using Remark 5.59) that

$$g(x) = \frac{a(x)}{b(x)}$$

is a continuous function $g : V \rightarrow \mathbb{R}$, where

$$a(x) = x^2 - 3x + 2, \quad b(x) = x^2 - 4x + 3,$$

and

$$V = \mathbb{R} \setminus \{1, 3\}.$$

Use Proposition 5.42 and Proposition 5.51 to show that

$$\{x \in \mathbb{R} \mid a(x) \leq 17\}$$

is a closed subset of $\mathbb{R}$.

Does

$$\lim_{x \rightarrow 3} g(x)$$

exist? What about

$$\lim_{x \rightarrow 1} g(x)?$$

1.1 Continuity of g

By Example 5.49 functions $f_1(x) = x$, $f_2(x) = 2$ and $f_3(x) = 3$ are continuous. By Proposition 5.57 function $a(x)$ given by $a(x) = f_1(x)f_1(x) - f_3(x)f_1(x) + f_2(x)$ is continuous and so is function $b(x)$ given by $b(x) = f_1(x)f_1(x) - f_2(x)f_2(x)f_1(x) + f_3(x)$. Then by proposition 5.57 function $g(x)$ given by $g(x) = \frac{a(x)}{b(x)}$ is continuous in domain V where $V = \{x \in \mathbb{R} \mid b(x) \neq 0\}$ which is the same as $V = \mathbb{R} \setminus \{1, 3\}$

1.2 Closed subset

Let I be set defined by $I = (-\infty, 17)$, let S be set $S = \{x \in \mathbb{R} \mid a(x) \in I\}$. By Definition 1.112 $S = a^{-1}(I)$. By Proposition 5.42 set I is closed and so is S by Proposition 5.51.

1.3 Existence of Limits

$$\begin{aligned} g(x) &= \frac{x^2 - 3x + 2}{x^2 - 4x + 3} \\ &= \frac{(x-1)(x-2)}{(x-1)(x-3)} \\ &= \frac{x-2}{x-3} \text{ in } V \end{aligned}$$

Limit as x approaches 1

$$\begin{aligned} \lim_{x \rightarrow 1} g(x) &= \lim_{x \rightarrow 1} \frac{x-2}{x-3} \\ &= \frac{1-2}{1-3} \\ &= \frac{1}{2} \end{aligned}$$

The limit as $x \rightarrow 1$ exists and has value 1

Limit as x approaches 3

$$\begin{aligned} \lim_{x \rightarrow 3} g(x) &= \lim_{x \rightarrow 3} \frac{x-2}{x-3} \\ &= \frac{1-2}{3-3} \\ &= \infty \end{aligned}$$

The limit as $x \rightarrow 3$ does not exist